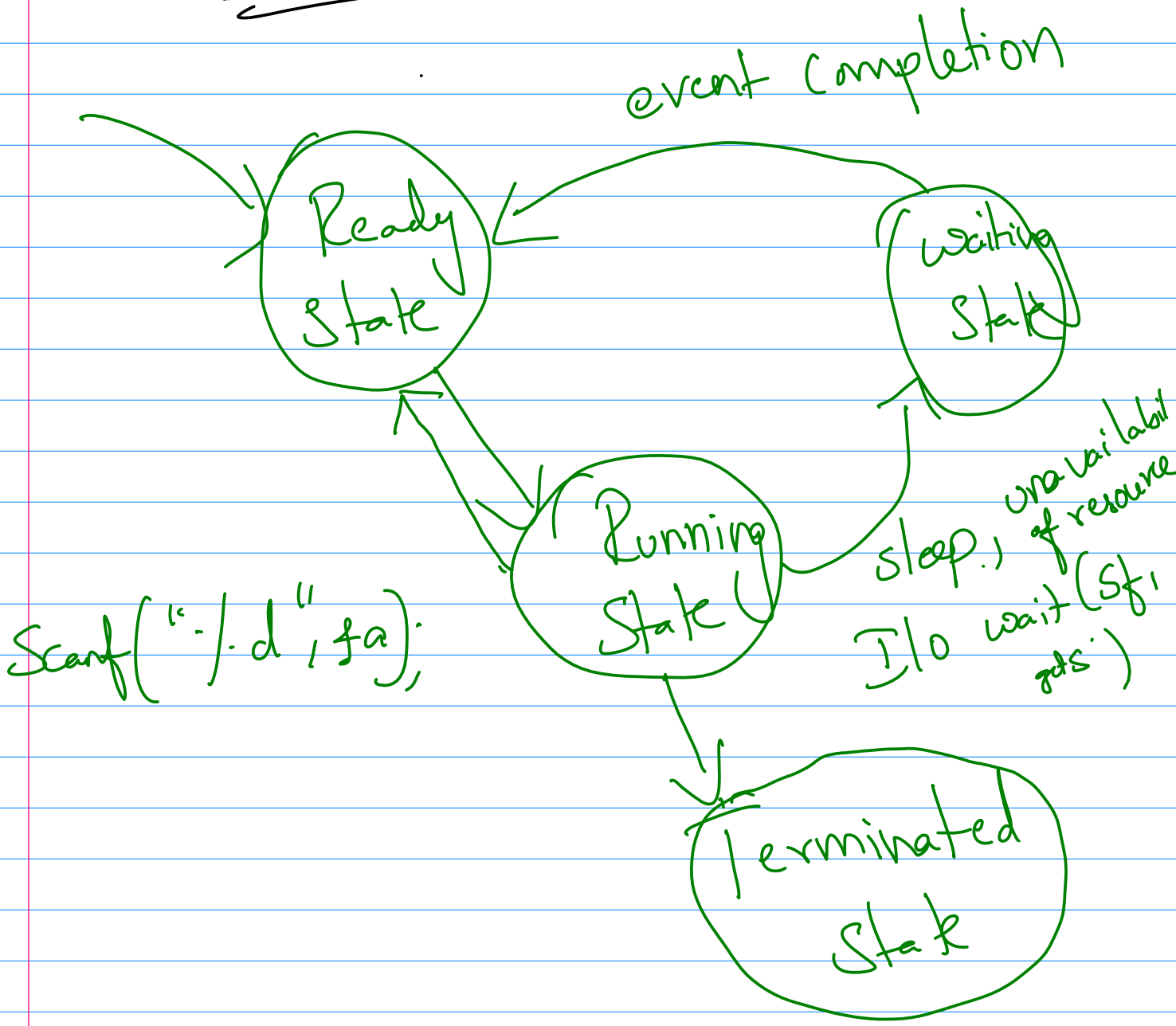
Process States

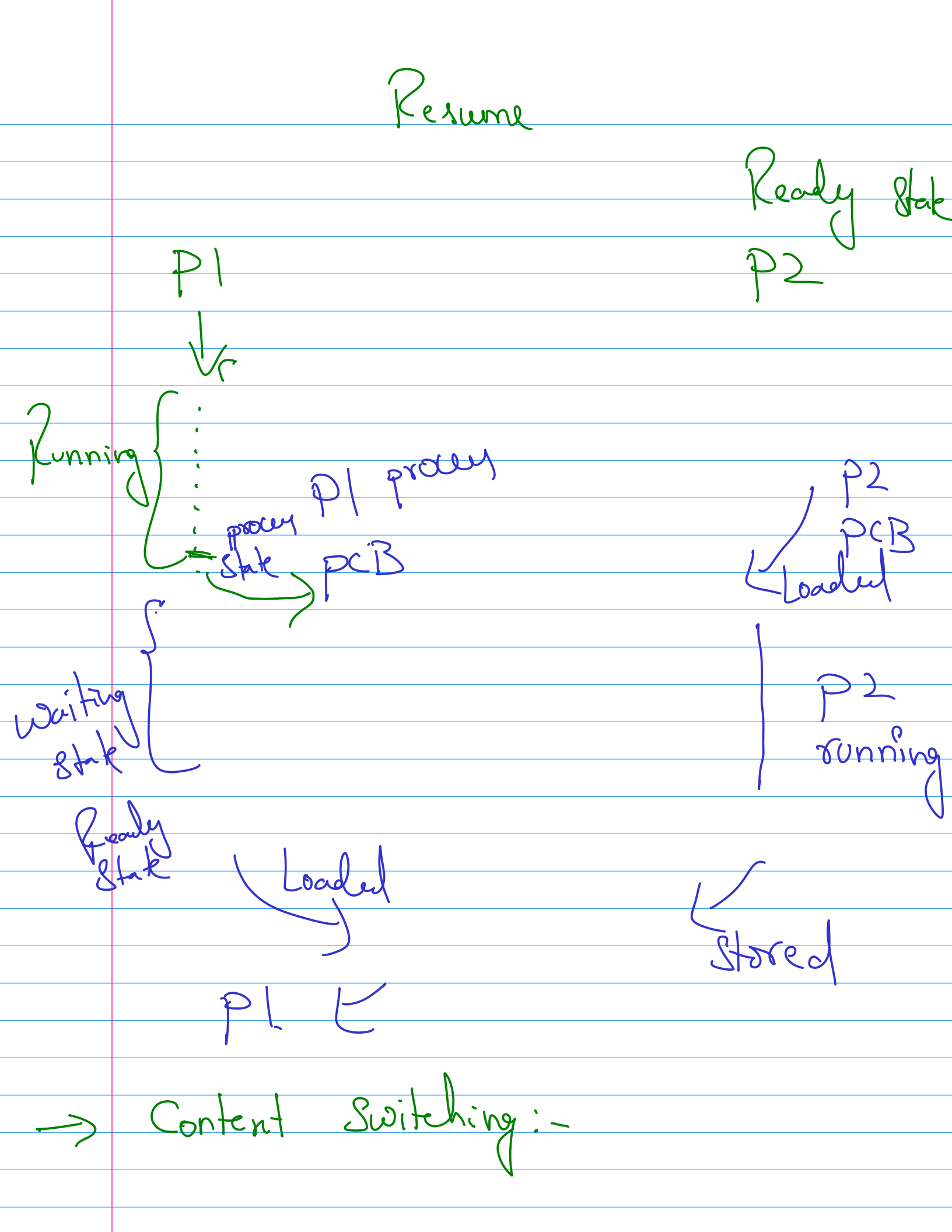


Draw the process state diagram when a below program one executes?

```
main()
{
    int a,b,c;
    printf("Enter the values for a & b \n");
    scanf("%d %d", &a, &b);
    c=a+b;
    sleep(10);
    printf("%d\n", c);
}
```

Ready State
↓
Running State
↓
Waiting state
↓
Ready State
↓
Running State
↓
Waiting state
↓
Ready State
↓
Running State
↳ Terminated State.

→ Suspended State
↳ only when Ready State.



→ Content Switching :-

→ storing the current proces inf^n into its PCB and loading the newly scheduled proces inf^n from the respective PCB is called Content Switching.

→ The time taken to do so Called as Content Switching time (CST) Dispatch latency

Q what is PCB & what it contains.

Ans:- process control block is a data struct maintained in the kernel space containing the inf^n about proces.

PCB Contains following infⁿ

process ID

parent pID

process state

CPU registers

SP

PC

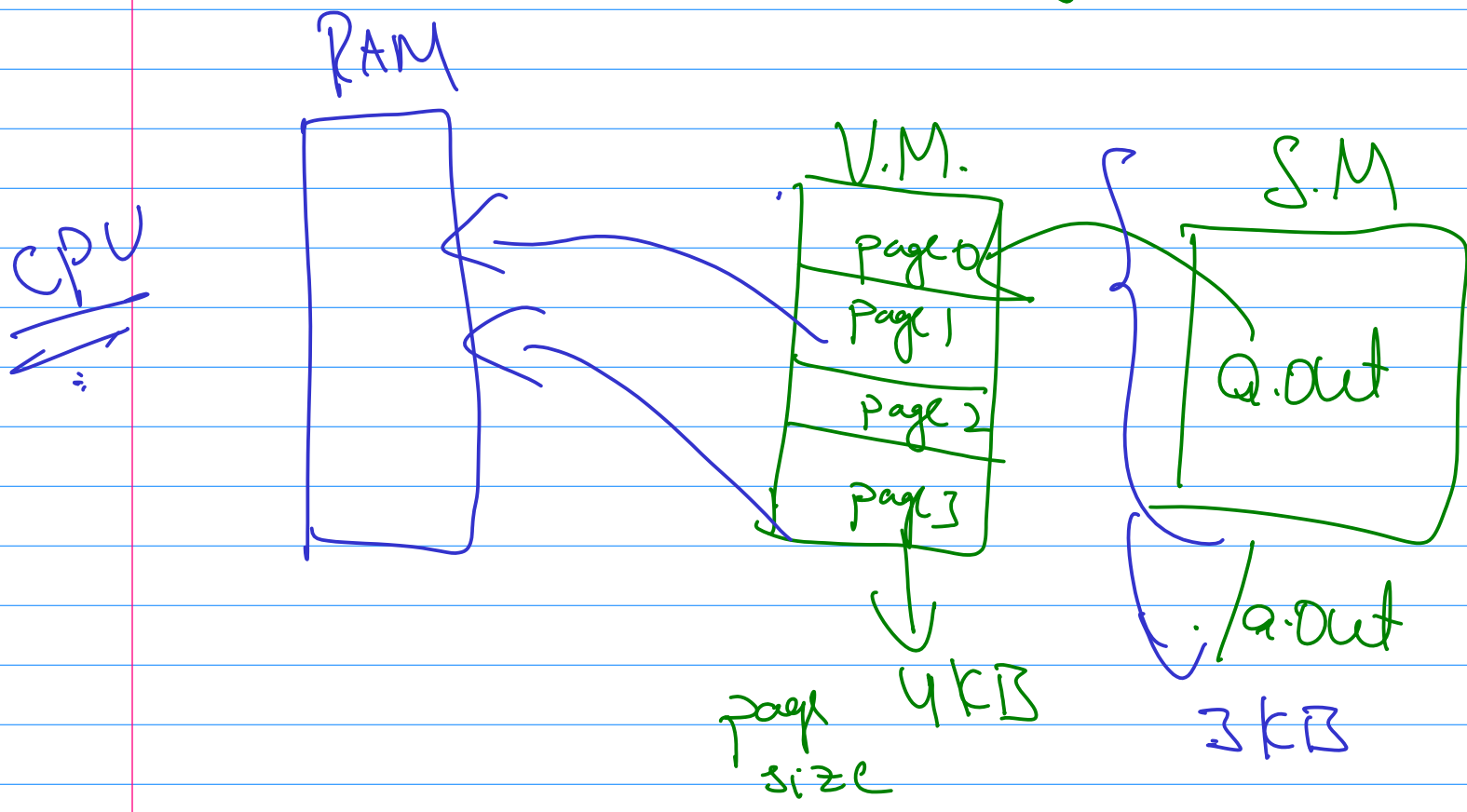
memory limits

ptr ↳ to the next PCB

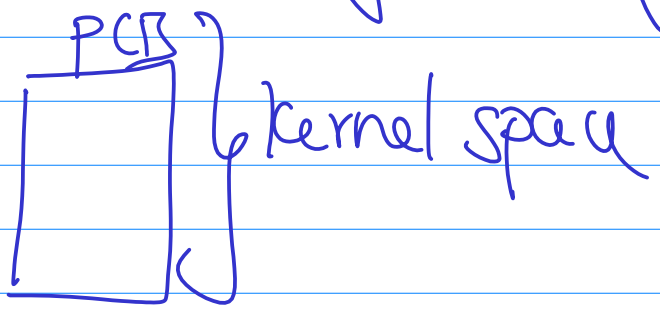
process entry point

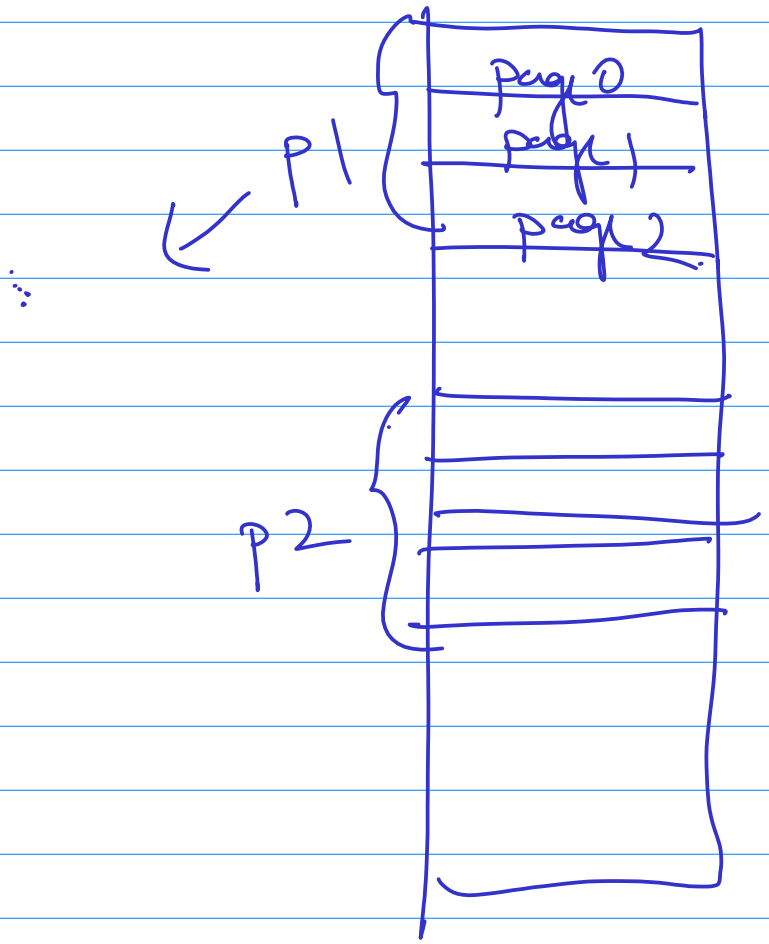
pages allocated to the process with address range

→ PID, PPID, Ready state, reg=default value,



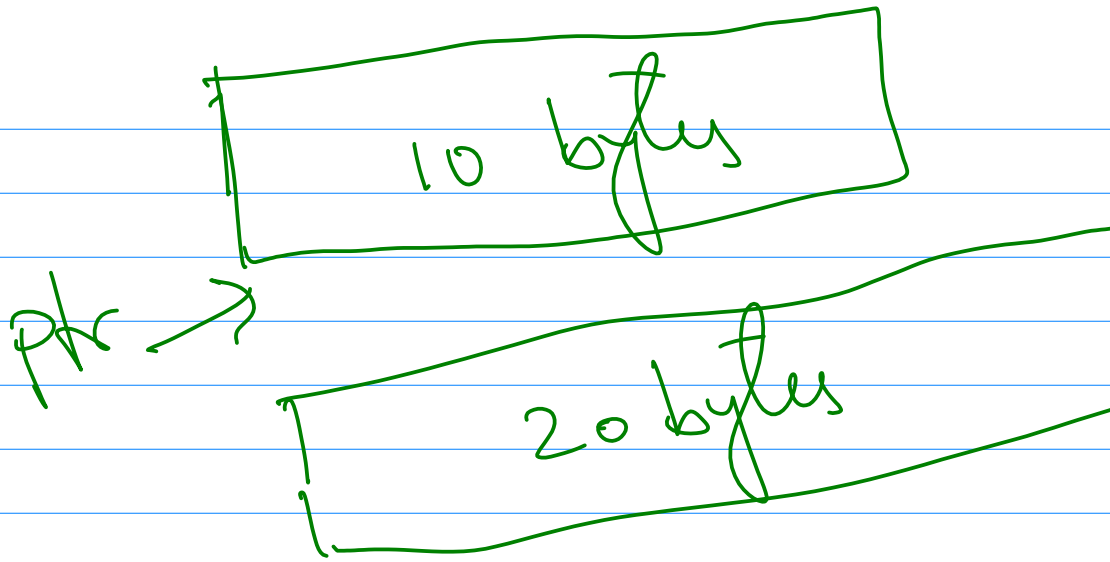
V-M → Secondary Memory





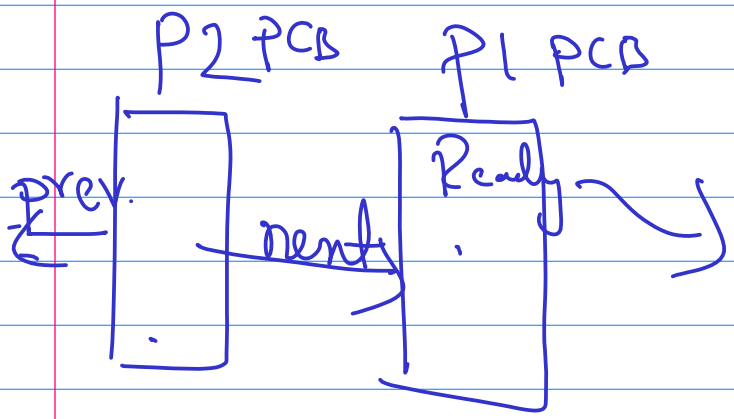
Q How free function how many bytes to be freed?

↳ gets the int^n pCB



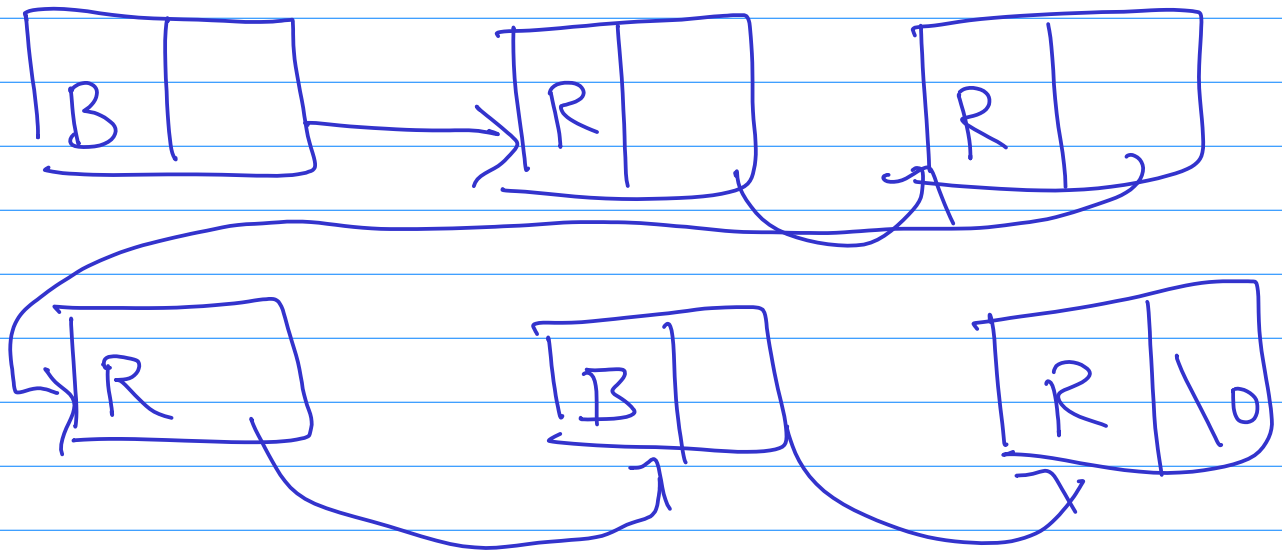
Ready State → processes in Ready state maintains a Queue
Ready Queue

Waiting State → Waiting Queue.



Seperate black nodes & red nodes

Q1



→ Waiting State → Ready State

Waiting Queue → Ready Queue

P1 → waiting

Sleep (100) → P2 ← P1

P2 → Sf → waiting

→ Input